
DistilBERT vs. BERT for Efficient Text Classification

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TODO: add institutional email if needed

Abstract

This mini-report studies the accuracy-efficiency trade-off between DistilBERT and BERT-base for text classification on AG News, SST-2, and Yelp Review Full. We fine-tune DistilBERT, run a DistilBERT-only ablation over classifier heads and freezing strategies, select the best DistilBERT configuration per dataset, and compare it with BERT-base. Evaluation uses accuracy, macro and weighted precision, recall, and F1, plus efficiency metrics including parameter count, latency, and peak GPU memory when available. The report is generated from saved experiment artifacts; debug-fast numbers are placeholders for pipeline validation and must be replaced by full-run results before drawing final conclusions.

1 Introduction

Text classification is a core natural language processing task with applications in topic classification, sentiment analysis, and review understanding. Transformer encoders such as BERT provide strong transfer learning performance by pretraining deep bidirectional representations [Devlin et al., 2019]. However, BERT-base is comparatively expensive in parameter count, memory, and inference latency, which can be limiting for deployment.

DistilBERT is a compressed Transformer model trained by knowledge distillation to retain much of BERT’s language understanding ability with fewer parameters and faster inference [Sanh et al., 2019]. This project asks whether DistilBERT can offer a favorable trade-off against BERT-base on three classification datasets: AG News, SST-2 from GLUE, and Yelp Review Full [Zhang et al., 2015, Wang et al., 2019].

The contributions are: (1) a reproducible data and training pipeline for DistilBERT and BERT-base; (2) a DistilBERT ablation over classifier size and freezing strategies; (3) a final comparison of the selected DistilBERT configuration against BERT-base; and (4) a compact analysis of performance and efficiency metrics.

2 Approach

All datasets are tokenized with the corresponding Hugging Face tokenizer and padded/truncated to a maximum sequence length of 128. The primary training objective is cross-entropy loss over dataset labels. For DistilBERT, the baseline keeps the Transformer encoder fully trainable and uses the standard classification head. The ablation study additionally evaluates a fully frozen Transformer, smaller and larger multilayer classifier heads, and a partial-freezing strategy that freezes embeddings and lower Transformer layers.

BERT-base uses `bert-base-uncased` with a comparable sequence classification head where possible. The model builder handles architecture-specific module names for DistilBERT and BERT, including embeddings and Transformer layer freezing. Models are evaluated using accuracy, macro precision/recall/F1, and weighted precision/recall/F1. Efficiency is measured using total parameters,

Table 1: Final performance comparison. Metrics are generated from results/tables/final_performance_table.csv.

Dataset	Model	Config	Acc.	F1	F1 w.
ag_news	distilbert	freeze_lower_layers	0.7500	0.4286	0.6429
ag_news	bert	bert_base	0.7500	0.4286	0.6429
sst2	distilbert	freeze_lower_layers	1.0000	1.0000	1.0000
sst2	bert	bert_base	0.7500	0.7333	0.7667
yelp_review_full	distilbert	small_classifier	0.2500	0.2000	0.4000
yelp_review_full	bert	bert_base	0.0000	0.0000	0.0000

Table 2: Final efficiency comparison. Parameter counts are shown in millions. Latency is in ms/sample.

Dataset	Model	Config	Params (M)	Trainable (M)	Latency	GPU MB
ag_news	distilbert	freeze_lower_layers	66.56	21.46	109.25	–
ag_news	bert	bert_base	109.49	109.49	245.26	–
sst2	distilbert	freeze_lower_layers	66.56	21.46	101.88	–
sst2	bert	bert_base	109.48	109.48	313.31	–
yelp_review_full	distilbert	small_classifier	66.46	66.46	116.33	–
yelp_review_full	bert	bert_base	109.49	109.49	320.41	–

trainable parameters, latency in milliseconds per sample, and peak GPU memory when CUDA measurements are available.

3 Experiments

The datasets cover topic classification (AG News), binary sentiment classification (SST-2), and five-class review rating prediction (Yelp Review Full). The default fine-tuning setup uses AdamW through the Hugging Face Trainer, learning rate 2×10^{-5} , weight decay 0.01, maximum length 128, and three epochs for full runs. Debug-fast mode uses tiny subsets and very few optimizer steps only to validate code paths.

The implementation uses PyTorch, Hugging Face Transformers, and Hugging Face Datasets [Paszke et al., 2019, Wolf et al., 2020, Lhoest et al., 2021]. Full-run experiments are intended for the target RTX 4090 environment with mixed precision enabled when CUDA is available. Debug-fast artifacts may appear in the generated tables if full-run results have not been produced; those values should not be interpreted as final experimental evidence.

4 Analysis

Table 1 summarizes final performance for the selected DistilBERT configuration and BERT-base. Table 2 reports efficiency metrics. Table 3 gives a compact ablation summary, and Table 4 lists the selected DistilBERT configuration per dataset.

The expected full-run analysis should compare whether BERT-base improves macro F1 enough to justify its larger parameter count and latency. If the selected DistilBERT configuration achieves similar F1 with fewer trainable parameters or lower latency, it is the more deployment-friendly model. If BERT-base materially improves accuracy or macro F1 on harder datasets such as Yelp Review Full, the additional compute may be justified for accuracy-critical settings. TODO: replace this paragraph with full-run findings after running the complete RTX 4090 experiments.

5 Conclusion

This report template completes the experimental pipeline by connecting saved metrics, plots, and tables into a NeurIPS-style mini-report. The current generated artifacts can validate the reporting

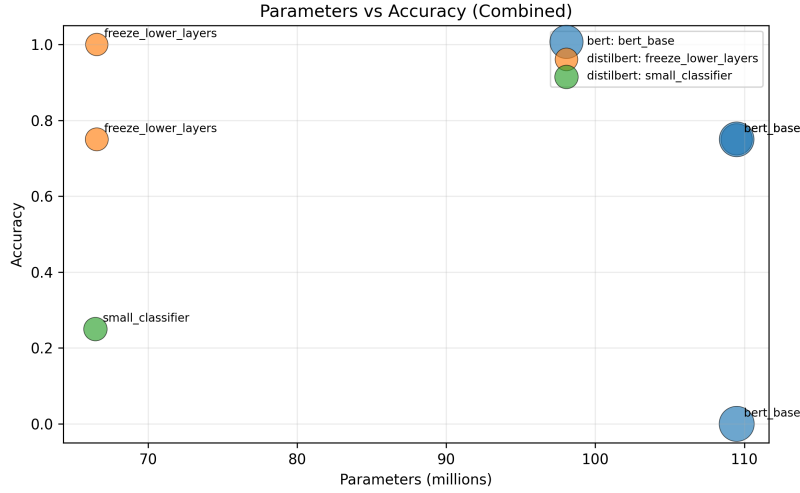


Figure 1: Accuracy versus parameter count. Bubble size uses latency when available, otherwise memory.

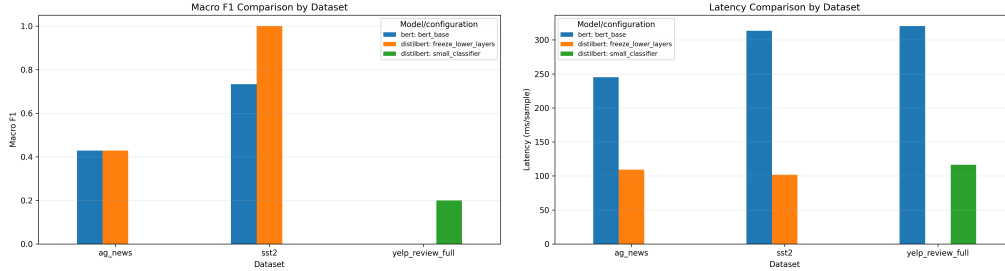


Figure 2: Macro F1 and latency comparison by dataset.

workflow, but final conclusions require full-run results. The final write-up should state the best DistilBERT configuration for each dataset, whether BERT-base improves accuracy or macro F1, and whether that improvement offsets the efficiency cost. Limitations include the small model family scope, fixed maximum sequence length, and limited hyperparameter search. Future work could evaluate additional compact encoders, calibration, throughput under batching, and statistical variation across random seeds.

References

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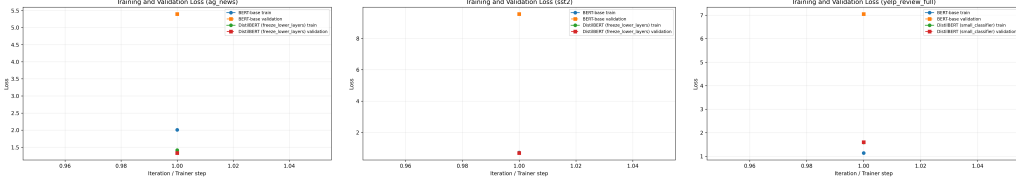


Figure 3: Training and validation loss curves for the available final comparison runs.

Table 3: DistilBERT ablation summary. To fit the mini-report format, the generated snippet keeps the top configurations per dataset.

Dataset	Config	Freeze	Head	Acc.	F1	Params (M)	Trainable (M)	Latency
ag_news	freeze_lower_layers	lower_layers	[256]	0.7500	0.4286	66.56	21.46	109.25
ag_news	baseline	none	[]	0.5000	0.2222	66.96	66.96	104.95
ag_news	frozen_transformer	all	[]	0.5000	0.2222	66.96	0.59	96.90
sst2	freeze_lower_layers	lower_layers	[256]	1.0000	1.0000	66.56	21.46	101.88
sst2	baseline	none	[]	0.5000	0.5000	66.96	66.96	111.12
sst2	frozen_transformer	all	[]	0.5000	0.5000	66.96	0.59	156.20
yelp_review_full	small_classifier	none	[128]	0.2500	0.2000	66.46	66.46	116.33
yelp_review_full	baseline	none	[]	0.0000	0.0000	66.96	66.96	218.68
yelp_review_full	freeze_lower_layers	lower_layers	[256]	0.0000	0.0000	66.56	21.46	94.28

Yang, Zachary DeVito, Martin Raison, Alykhan Tejani, Sasank Chilamkurthy, Benoit Steiner, Lu Fang, Junjie Bai, and Soumith Chintala. Pytorch: An imperative style, high-performance deep learning library. In *Advances in Neural Information Processing Systems*, volume 32, 2019.

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Table 4: Selected DistilBERT configuration per dataset.

Dataset	Best config	Acc.	F1	Params (M)	Trainable (M)	Latency	GPU MB
ag_news	freeze_lower_layers	0.7500	0.4286	66.56	21.46	109.25	–
sst2	freeze_lower_layers	1.0000	1.0000	66.56	21.46	101.88	–
yelp_review_full	small_classifier	0.2500	0.2000	66.46	66.46	116.33	–